

Nour Waleed Fahmy

Data Cleaning & ML Analysis.

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Profile

Final-year CS student (AI minor) who **cleans, analyses and models data** in Python. Hands-on with imbalanced classification, feature engineering and regression. Also built production backend work in Node.js and Express with MySQL, focused on reliable APIs, security, and documentation.

Core Skills

Programming



C++, Java, Python, PHP, TypeScript, SQL, C, HTML, CSS, JavaScript

Automation & Backend



Node.js, REST APIs, Express, Swagger (API docs), MongoDB, SQL (MySQL/XAMPP), Workbench

Data & ML tools



pandas, NumPy, scikit-learn, XGBoost, imbalanced-learn, matplotlib, seaborn, Excel, Google Sheets, Orange (data mining), Power BI

Engineering Tools



GitHub, Bitbucket, Linux (Ubuntu), VS Code, Jupyter / Colab, XAMPP, Heroku

Selected Coursework

Machine Learning | Image Processing | Data Structures & Algorithms | SWE Project Management | Computer Networks | Database Management Systems | Web Development | OOP

Projects

Diabetes Prediction Model Comparison (team project)

- Cleaned a 100k-row dataset (removed 3,854 duplicate rows) with a positive class of only 8.8%
- Compared several model families on balanced accuracy, recall, precision and F1; best model reached 0.896 balanced accuracy and 86.6% recall on a held-out test set

Laptop Price Prediction

- Engineered features from raw specs (screen density, storage type, CPU generation) and compared six regression approaches
- Best validation result: R^2 0.91, mean absolute error about €150 (Ridge, log-transformed target)

Hotel Booking Cancellation Prediction

- Trained and compared KNN, Decision Tree, and Neural Network models to predict cancellations
- Handled imbalance using SMOTE and applied Genetic Algorithm feature selection

Image Processing

- Detected walking pedestrians using MOG2 background subtraction on video frames
- Tuned parameters and post-processing to reduce noise and false detections

Additional Projects

Automated Scheduling System (Laravel, Tailwind), Campus Network (Packet Tracer), Raising Roots (Node.js, MongoDB), Recruitment System (JavaFX)

Experience

Backend Intern | Step In (Internship Program Project)

07/2025 – 08/2025

webkeyz

Cairo, Egypt

- Worked on the backend using Node.js and Express with JavaScript or TypeScript, backed by MySQL
- Built secure REST APIs with JWT and session based authentication plus role based access control
- Delivered core modules including file uploads, admin APIs, notifications, and search and filtering, documented with Swagger and validated via Postman

Event Usher | John Gray Event

Nexus

- Supported on site operations and attendee assistance, helping keep the event organized and running smoothly

Education

B.Sc. in Computer Science (AI minor)

2023 – present

Misr International University (MIU)

Obour, Egypt

High School Diploma

2023

Amgad Language School

Cairo, Egypt

Languages

Arabic



English



French

